

Information Retrieval & Data Pipeline Development

From Collection to Analysis

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Course Structure

Three Main Components

- **Data Acquisition:** Web scraping techniques
- **Data Storage:** Multiple database systems
- **Data Analysis:** From EDA to ML modeling

Project-Based Learning

Build complete pipeline: scrape → store → analyze

Web Scraping Basics

- Send HTTP requests to fetch web pages
- Parse HTML content using tag-based structure
- Extract specific elements using tag names, classes, or IDs
- Convert extracted data to CSV or DataFrame

Multi-Format Storage

CSV (Pandas)

```
[language=Python]  
df.to_csv('data.csv')
```

SQL (SQLite)

```
[language=SQL]  
CREATE TABLE  
products ( id INTEGER  
PRIMARY KEY, name  
TEXT, price REAL );
```

NoSQL (MongoDB)

```
[language=Python]  
db.products.insert_one("na
```

Exploratory Data Analysis (EDA)

Python Example

```
import pandas as pd
df = pd.read_csv('data.csv')
Basic stats
df.describe()
Visualization
df['price'].hist()
```

- Descriptive statistics
- Frequency distributions
- Correlation analysis
- Outlier detection

- Handling missing values
- Encoding categoricals
- Creating new features
- Time-based features

Common Techniques

Normalization

$df['norm_price'] = (df['price'] - df['price'].mean()) / df['price'].std()$

One-hot encoding `pd.get_dummies(df['category'])`

Predictive Analysis

- Regression/Classification
- Model evaluation
- Hyperparameter tuning
- Feature importance

Machine Learning Pipeline

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y)
model = RandomForestRegressor()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

End-to-End Project

- 1 Identify data source and scrape
- 2 Design database schemas
- 3 Store in CSV, SQL, and NoSQL
- 4 Perform EDA and visualization
- 5 Feature engineering
- 6 Build predictive models
- 7 Document findings

Example Project

Scrape e-commerce product data → Price prediction model

Python Libraries

- BeautifulSoup / Scrapy
- Pandas / NumPy
- Matplotlib / Seaborn
- Scikit-learn
- SQLAlchemy / PyMongo

Databases

- SQLite / PostgreSQL
- MongoDB / Firebase
- Redis (caching)

Containerization

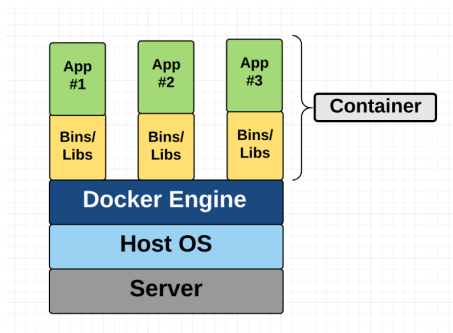
• **Docker:**

- Utilize database and tools
- Reproducible environment setup
- Easy deployment across systems

Introduction to Docker

What is Docker?

- Docker is a containerization platform.
- It allows packaging of applications and their dependencies into portable containers.
- Containers are lightweight and consistent across environments.
- Simplifies reproducibility of data pipelines and ML models.



<https://t.ly/MqmVD>

Thank You!